

Federated Learning for Chest X-Ray Classification Across Geographically Distributed Hospitals Without Data Sharing

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ABSTRACT

Training diagnostic AI on large centralised datasets raises significant data governance, privacy, and regulatory concerns. Federated learning (FL) offers a solution: models are trained locally at each site and only model gradients — never patient data — are shared. We conducted the largest prospective FL study in medical imaging to date, training a chest X-ray classifier across 23 hospitals in 9 countries without a single patient image leaving its originating institution. The resulting model achieves AUC 0.91 for 14 pathology classes, comparable to a model trained on the same data centrally (AUC 0.93). We demonstrate that FL preserves model performance while reducing privacy risk and eliminating cross-border data transfer concerns.

METHODS

- Participants: 23 hospitals, 9 countries, 890,000 chest X-rays
- FL architecture: FedAvg with differential privacy noise injection ($\epsilon = 3.0$)
- Communication rounds: 150; local epochs per round: 5
- Pathology classes: 14 (pneumonia, effusion, cardiomegaly, etc.)
- Comparison: centralised training on same dataset as gold standard
- Privacy audit: formal ϵ -differential privacy guarantee documented

KEY FINDINGS

1. FL model AUC 0.91 vs 0.93 for centralised — performance gap minimal
2. Federated training reduced communication cost by 78% vs baseline FL approaches
3. Differential privacy added only 2.1% performance reduction
4. Hospital heterogeneity (different scanners, patient populations) handled by personalised layers
5. Regulatory compliance achieved in all 9 jurisdictions without bespoke data agreements

LIMITATIONS

- Communication rounds require stable internet — challenging in low-resource settings
- Convergence slower with high between-site heterogeneity
- Differential privacy noise impacts small-site performance disproportionately
- Model update poisoning attacks not tested in adversarial setting

RELEVANCE TO ML IN CLINICAL DIAGNOSTICS: HIGH

Directly addresses the primary barrier to large-scale clinical ML deployment: data governance. The multi-jurisdiction approach is a significant practical contribution.